TIMING NEUTRON STARS

and the link with black-hole timing

M. van der Klis
Astronomical Institute "Anton Pannekoek"
University of Amsterdam
michiel@science.uva.nl

Abstract This lecture deals with aspects of the rapid X-ray variability of low-magnetic neutron stars in low-mass X-ray binaries. After summarizing some basic considerations related to the physics of orbital motion in strong field gravity and introducing the main types of system, the relations of the kilohertz quasi-periodic oscillations with orbital motion, neutron star spin, luminosity and accretion rate and other, slower variability components are reviewed. Finally, possible relations with the variability of black holes are discussed.

Keywords: stars:neutron – stars:pulsar – X-rays:binaries – X-rays:stars – black hole physics – stars:oscillations

1. Introduction

X-ray binaries are interesting in their own right and from the point of view of stellar evolution, but perhaps their main interest is in their role as cosmic laboratories to study strong gravity and dense matter. Two prominent aims of X-ray binary studies are the direct determination of the properties of the strong gravitational fields near black holes and neutron stars, and of the supranuclear density matter in the interior of neutron stars. In the inner few kilometers of the accretion flow onto a low-magnetic-field neutron star or stellar-mass black hole of mass M the accreting matter is moving close to the Schwarzschild radius at $R_S = 2GM/c^2$. This means we are looking at the motion of matter in strongly curved spacetime and can learn about gravitation in the strong-field regime, a regime where classical physics fails and general relativity has not yet been tested to any degree of confidence by means other than extrapolation from observations in weak gravity. In the case of a neutron star, the properties of the accretion flow constrain fundamental neutron star parameters such as mass and radius, and thereby (via stellar structure theory) the equation of state

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(EOS) of the ultradense matter at the star's core, whose properties are the subject of much speculation involving, among others, pion and kaon condensates, hyperons, and even strange matter. Measurements of the varying emission of hot spots on the surface of a spinning neutron star can likewise constrain both gravitation theory and neutron star parameters.

X-ray timing as well as spectroscopy can be used to study the dynamics of the inner accretion flows and the neutron star spins; for best results one wants to combine these techniques. Both X-ray binaries and AGNs are useful to study the accretion flow in the strong-field region; both types of system have advantages, and again for best results one wants to study both, and describe the results in one coherent picture. An important aspect of compact object studies that is unique to X-ray binaries is the possibility to compare systems that are similar except in the type of compact object they contain. A particularly interesting comparison is that between low-magnetic field neutron stars and black holes, which are both expected to allow gravity-dominated accretion flows down to well into the strong-field region. This comparison, at least in principle, allows to determine which of the observed properties are caused by the unique characteristics of a black hole, and which are in common with other accreting compact objects. Future additional techniques will allow to address these issues in new ways. One is direct imaging of the strong-field region, which becomes possible (in principle) initially in radio to IR/sub-mm wavelengths for the galactic center and AGNs, and only later in X-rays, as the latter requires the development of large baseline X-ray interferometers, which is very challenging indeed. Another is gravitational wave detection of accretion and merging events involving compact objects; ground based gravitational wave experiments such as LIGO and later space based ones such as LISA are directed towards this aim.

My charge for this review is to concentrate on neutron-star X-ray binary timing. I shall put particular emphasis on the millisecond time variability of the X-rays emitted by the accreting low-magnetic field neutron stars in low mass X-ray binaries, as it is in these neutron stars that a plasma flow not dominated by the neutron star magnetic field extends down into the strong gravity region. The link of the timing phenomena observed in these neutron stars with those in stellar mass black holes, where the same basic physical situation applies, is also addressed. The emphasis on *millisecond* variability comes from the consideration that the dynamical time scales of stellar mass compact objects are of this order; hence the main characteristic time scales in the strong-field region, such as the orbital time scale, as well as the spins of weakly magnetic neutron stars spun up by accretion, tend to be of the order of milliseconds as well. I shall be very brief in this write-up on the topic of the data analysis of rapid X-ray variability and just mention that, certainly in the millisecond domain the analysis method of choice for detecting the variability and measuring